

DYNAMICAL SYSTEMS - MA3081

Solutions to Exercise Sheet 3

Exercise 3.1 (Invariant Sets 2).

Consider an invertible continuous dynamical system $(\mathcal{X}, \mathcal{T}, \phi_t)$.

- a) Let $x_0 \in \mathcal{X}$. Show that the ω -limit set $\omega(x_0)$ is invariant.
- b) Let S be an invariant set. Show that the closure, the interior and the boundary of S are positively invariant.
- c) Let $S_1, \dots, S_n$ be invariant sets. Show that the intersection and the union of $S_1, \dots, S_n$ are again invariant.

Solution.

- a) Let $\hat{x} \in \omega(x_0)$ and let $s \in \mathcal{T}$. By definition there exists a sequence $t_n \rightarrow \infty$ with $\phi_{t_n}x_0 \rightarrow \hat{x}$ as $n \rightarrow \infty$. By the properties of a continuous dynamical system, we have $\phi_{s+t_n}x_0 = \phi_s \circ \phi_{t_n}x_0 \rightarrow \phi_s\hat{x}$ as $n \rightarrow \infty$. Therefore $h_n := s + t_n$ defines a sequence with $h_n \rightarrow \infty$ as $n \rightarrow \infty$ and $\phi_{h_n}x_0 \rightarrow \phi_s\hat{x}$. By definition that means that $\phi_s\hat{x} \in \omega(x_0)$. As $s \in \mathcal{T}$ was arbitrary, this holds for every $s \in \mathcal{T}$. Since $\hat{x}$ was arbitrary as well, we have $\phi_s\omega(x_0) \subset \omega(x_0)$ and thus $\omega(x_0)$ is invariant.
- b) Let $\bar{S}$ denote the closure of S and let $\hat{x} \in \bar{S}$. Then, there is a sequence $x_n \in S$ with $x_n \rightarrow \hat{x}$ as $n \rightarrow \infty$. Now, let $t \in \mathcal{T}$ and consider the sequence $\phi_t x_n$. By invariance of S , we have $\phi_t x_n \in S$. Moreover, $\phi_t(x_n) \rightarrow \phi_t\hat{x}$ by continuity. Consequently, $\phi_t\hat{x} \in \bar{S}$ for every $\hat{x} \in \bar{S}$ and every $t \in \mathcal{T}$. This proves that $\bar{S}$ is invariant.
To prove that the interior S° of S is invariant, note that $S^\circ = \mathcal{X} \setminus \overline{\mathcal{X} \setminus S}$. Invariance of S° follows from Exercise 1.4a).
To show that the boundary ∂S is invariant, note that $\partial S = \bar{S} \setminus S^\circ$. Thus, invariance follows by a similar argument.
- c) Let $x \in \cup_{i=1}^n S_i$. Thus, $x \in S_k$ for some k . By invariance of S_k , we have $\phi_t x \in S_k$ for all $t \in \mathcal{T}$. Thus, $\phi_t x \in \cup_{i=1}^n S_i$ and the union of $S_1, \dots, S_n$ is invariant. The invariance of the intersection can be shown analogously.

□

Exercise 3.2 (Lyapunov function).

Consider the following system of differential equations

$$\begin{aligned}x' &= y, \\y' &= -2x - y - 3x^5.\end{aligned}$$

Show that the origin is a stable equilibrium point using a Lyapunov function of the form $L(x, y) = \alpha x^2 + \beta y^2 + \gamma x^6$ for suitable coefficients $\alpha, \beta, \gamma \in \mathbb{R}$.

Solution.

We seek a Lyapunov function of the form $L(x, y) = \alpha x^2 + \beta y^2 + \gamma x^6$ with some $\alpha, \beta, \gamma \in \mathbb{R}$. First note that we always have $L(0, 0) = 0$ and further $L(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2 \setminus \{0\}$ if we choose $\alpha, \beta, \gamma \geq 0$.

We compute the Lie derivative

$$\begin{aligned}\nabla L(x, y) \cdot f(x, y) &= \begin{pmatrix} 2\alpha x + 6\gamma x^5 \\ 2\beta y \end{pmatrix} \cdot \begin{pmatrix} y \\ -2x - y - 3x^5 \end{pmatrix} \\ &= 2\alpha xy + 6\gamma x^5 y - 4\beta xy - 2\beta y^2 - 6\beta x^5 y \\ &= (2\alpha - 4\beta)xy + (6\gamma - 6\beta)x^5 y - 2\beta y^2\end{aligned}$$

If we now choose α, β, γ in such a way that

$$2\alpha = 4\beta, \quad 6\gamma = 6\beta,$$

for example

$$\alpha = 2, \quad \beta = 1, \quad \gamma = 1$$

this simplifies to

$$\nabla L(x, y) \cdot f(x, y) = -2\beta y^2 = -2y^2 \leq 0.$$

Theorem 2.4 now allows to conclude that the equilibrium at the origin is stable. $\square$

Exercise 3.3 (Stable/unstable manifolds of the Duffing equation).

Consider the system

$$\begin{aligned}\dot{x} &= y \\ \dot{y} &= x - px^3\end{aligned}\tag{3.1}$$

with some parameter $p \in \mathbb{R}$, that arises out of the unforced duffing equation $\ddot{x} + \delta \dot{x} - x + px^3 = 0$ with $\delta = 0$ (see also Example 3.10 from the lecture).

- Find the equilibria and check if they are hyperbolic.
- Compute a polynomial approximation of the stable and unstable manifolds of the equilibrium $(0, 0)$.
- Exploit the special structure of (3.1) to find an explicit expression of these manifolds. What do you observe when you extend the local stable/unstable manifolds to their global counterpart?

Solution.

- Equilibria at $(0, 0)$ and $(\pm \frac{1}{\sqrt{p}}, 0)$ if $p > 0$. $(0, 0)$ is a hyperbolic saddle with eigenvalues -1 and 1 , while $(\pm \frac{1}{\sqrt{p}}, 0)$ is not hyperbolic.
- We use the fact that the stable and unstable manifolds are uniquely characterized by invariance and tangency to the corresponding eigenspaces. First, we find the eigenspaces as $E_u = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$ and $E_s = \text{span} \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$.

Now one can proceed with a coordinate change so that the new coordinates represent the eigendirections, but we will skip this step here.

Since the eigendirections are not orthogonal to the x -direction we can not only obtain the stable/unstable manifold as a graph over the corresponding eigenspace, but also as a graph over x . (In principle also a graph over y is possible, but taking the x -direction yields the easiest computations here.)

We make an ansatz $y = h(x) = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^3 + \dots$ for the manifolds. Since we require $y(0) = 0$ (so that $(0, 0) \in W_{loc}^{s,u}$), we immediately have $\alpha_0 = 0$. And since the stable and unstable manifolds are tangent to their corresponding eigenspaces $E_{s,u}$, we must have $\alpha_1 = 1$ for the unstable manifold and $\alpha_1 = -1$ for the stable manifold.

To enforce invariance of the graph of h , we consider $\mathcal{F}(x, y) := h(x) - y$. Clearly, $\mathcal{F}(x, y) = 0 \iff (x, y)$ lies on the graph. Furthermore, the graph of h is invariant iff the Lie derivative of $\mathcal{F}$ vanishes on the entire graph. We have

$$\frac{d}{dt}\mathcal{F}(x, y) = h'(x)\dot{x} - \dot{y} = h'(x)y - (x - px^3).$$

We set this equal to 0 and insert $y = h(x)$ to arrive at the invariance equation

$$x - px^3 = h'(x) \cdot h(x).$$

We insert the representation of h into the invariance equation to obtain

$$\begin{aligned} x - px^3 &= (\alpha_1 + 2\alpha_2 x + 3\alpha_3 x^2 + \dots) (\alpha_0 + \alpha_1 x + \alpha_2 x^2 + \alpha_3 x^3 + \dots) \\ \Rightarrow x - px^3 &= (\pm 1 + 2\alpha_2 x + 3\alpha_3 x^2 + \dots) (\pm x + \alpha_2 x^2 + \alpha_3 x^3 + \dots) \\ \Rightarrow x - px^3 &= x + (\pm\alpha_2 \pm 2\alpha_2)x^2 + (\pm\alpha_3 + 2\alpha_2^2 \pm 3\alpha_3)x^3 + \dots \end{aligned}$$

Comparing the coefficients of x, x^2, x^3 yields

$$\begin{aligned} \alpha_2 &= 0, \\ \alpha_3 &= \mp \frac{p}{4}, \end{aligned}$$

so that the unstable manifold is approximately given as the graph of $h(x) = x - \frac{p}{4}x^3$, while the stable manifold is approximately given as the graph of $h(x) = -x + \frac{p}{4}x^3$.

- c) (3.1) is a Hamiltonian system with $H(x, y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{p}{4}x^4$. Since the stable/unstable manifold converge to the equilibrium in forward/backward time they have to lie on the same level set. We have $H(0, 0) = 0$ so also the stable/unstable manifold must lie on the zero level set. Hence they solve the equation $\frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{p}{4}x^4 = 0$. This equation has the solution $y = \pm \sqrt{x^2 - \frac{p}{2}x^4}$ Taylor expanding this around $x = 0$ yields $y = \pm \sqrt{x^2 - \frac{p}{2}x^4} = \pm x \mp \frac{p}{4}x^3 + \dots$ in accordance to b). We can also observe that the stable and unstable set are connected and thus are the same set.

□

Exercise 3.4 (Stable manifolds for maps).

Let $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$g \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y^2 \\ 2y - 2x^2 + x^4 y^2 \end{pmatrix}$$

and consider the dynamical system $(\mathbb{R}^2, \mathbb{N}_0, \phi_t)$ that is induced by iterating the map g .

- a) Find all fixed points of g .
- b) Show that $S = \{y = x^2\}$ is an invariant set.
- c) Consider the fixed point $(0, 0)$. It is hyperbolic? If so, compute its stable and unstable eigenspaces.
- d) Conclude that $W_{\text{loc}}^s(0, 0) = \{(x, y) \in \mathbb{R}^2 : (x, y) \in S \text{ and } |x| < \varepsilon\}$ for some $\varepsilon > 0$.

Solution.

- a) To find fixed points, we have to solve $g(x, y) = (x, y)$. This yields the equations

$$x = y^2, \tag{3.2}$$

$$y = 2y - 2x^2 + x^4y^2. \tag{3.3}$$

Inserting (3.2) into (3.3) yields $y - 2y^4 + y^{10} = 0$. One can immediately see that $y_1 = 0$ is a solution, so it remains to study $1 - 2y^3 + y^9 = 0$. Introducing $z = y^3$ this is equivalent to $1 - 2z + z^3 = 0$ and one can guess the solution $z = 1$, which corresponds to $y_2 = 1$. After a polynomial division, one is left with $z^2 + z - 1 = 0$, which has solutions $z_{1,2} = (-1 \pm \sqrt{5})/2$ that correspond to $y_3 = \sqrt[3]{z_1}$ and $y_4 = \sqrt[3]{z_2}$. This yields four fixed points

$$(0, 0), \quad (1, 1), \quad (y_3^2, y_3), \quad (y_4^2, y_4).$$

- b) Let $(\hat{x}, \hat{y}) \in S$, i.e. $\hat{x}^2 = \hat{y}$. Then, $g(\hat{x}, \hat{y}) = g(\hat{x}, \hat{x}^2) = (\hat{x}^4, 2\hat{x}^2 - 2\hat{x}^2 + \hat{x}^4\hat{x}^4) = (\hat{x}^4, \hat{x}^8) = (\bar{x}, \bar{x}^2)$ for $\bar{x} = \hat{x}^4$. Thus, $g(\hat{x}, \hat{y}) \in S$.
- c) Linearizing g at the origin yields

$$D_{(x,y)}g(0, 0) = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$$

Thus, the multipliers of $(0, 0)$ are given by 0 and 2. Therefore, $(0, 0)$ is hyperbolic and the stable and unstable eigenspaces are given by $E^s(0, 0) = \{(x, y) \in \mathbb{R}^2 : y = 0\}$ and $E^u(0, 0) = \{(x, y) \in \mathbb{R}^2 : x = 0\}$.

- d) Since the fixed point is hyperbolic, one can apply the stable/unstable manifold theorem, which guarantees the existence of a one-dimensional unstable and a one-dimensional stable manifold near $(0, 0)$. Both of them are tangent to the respective unstable and stable eigenspaces. Since S is a one-dimensional invariant manifold, that is tangent to the stable eigenspace, it locally has to be the stable manifold.

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